

# Calculus: Derivatives, Differentiation Rules & Tangent Lines

Grade 11–12 · Worked examples and practice reference, typeset from class notes

## Topics covered in this session

1. Graphical representation of a derivative · 2. Tangents and normals · 3. Maxima and minima (previewed)

### 1. The derivative as a slope

For two points  $(x_1, y_1)$  and  $(x_2, y_2)$  on a line, the slope  $m$  equals the tangent of the angle  $\theta$  the line makes with the  $x$ -axis:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \tan \theta$$

For a curve  $y = f(x)$ , the derivative is defined as the limit of the average rate of change as the interval  $h$  shrinks to zero:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

This is exactly the same idea as slope — the derivative at a point gives the slope of the curve (i.e. of its tangent line) at that point:

$$f'(x) = \frac{y_2 - y_1}{x_2 - x_1} = m = \text{slope}$$

### 2. Differentiation rules

i  $\frac{d}{dx}(x^n) = nx^{n-1}$

*Power rule.*

ii  $\frac{d}{dx}(c) = 0$ ,  $c$  is a constant

*Derivative of a constant.*

iii  $\frac{d}{dx}(uv) = v \frac{du}{dx} + u \frac{dv}{dx}$

*Product rule.*

$$\text{iv} \quad \frac{d}{dx} \left( \frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

Quotient rule.

Chain rule — for a composite function  $h(x) = (u \text{ composed with } v)(x)$ :

$$h(x) = (u \circ v)(x)$$

$$\text{v} \quad h'(x) = \frac{d}{dx} [(u \circ v)(x)] = \frac{d}{dx} [u(v(x))] \cdot \frac{d}{dx} [v(x)]$$

$$\text{vi} \quad y = e^x \Rightarrow y' = e^x$$

Exponential rule.

### Trigonometric derivatives

$$\text{vii} \quad \frac{d}{dx} (\sin x) = \cos x$$

$$\text{viii} \quad \frac{d}{dx} (\cos x) = -\sin x$$

$$\text{ix} \quad \frac{d}{dx} (\tan x) = \sec^2 x$$

$$\text{x} \quad \frac{d}{dx} (\sec x) = \sec x \tan x$$

$$\text{xi} \quad \frac{d}{dx} (\cot x) = -\csc^2 x$$

$$\text{xii} \quad \frac{d}{dx} (\csc x) = -\csc x \cot x$$

## 3. Separation of constants

When differentiating  $f(ax)$ , the constant  $a$  is carried straight through:

$$\frac{d}{dx} f(ax) = a f'(ax)$$

Example:

$$\frac{d}{dx} \sin(7x) = 7 \cos(7x)$$

## 4. Worked examples

(a)

$$y = x^5 \Rightarrow \frac{dy}{dx} = 5x^4$$

**(b) — a pure constant differentiates to zero**

$$y = c^5 \text{ (c constant)} \Rightarrow \frac{dy}{dx} = 0$$

**(c) — product rule**

$$y = x^3(1 + x)$$
$$y' = (1 + x)(3x^2) + 1 \cdot x^3 = 3x^3 + 3x^2 + x^3 = 4x^3 + 3x^2$$

**(d) — quotient rule**

$$y = \frac{1+x}{x^2}$$
$$y' = \frac{x^2(1) - (1+x)(2x)}{x^4} = \frac{x - 2 - 2x}{x^3} = -\frac{x+2}{x^3}$$

**(e) — chain rule**

$$y = (1+x)^5$$
$$y' = 5(1+x)^4 \cdot \frac{d}{dx}(1+x) = 5(1+x)^4$$

## 5. Tangent lines

A **tangent** is a line that touches a curve at exactly one point. For  $y = x^2$ , the derivative is  $2x$ , so the slope of the tangent at any point  $x$  is given by:

$$y = x^2 \Rightarrow \frac{dy}{dx} = 2x$$

### Point-slope form of a tangent

The equation of the tangent to a curve at the point  $(x_1, y_1)$  is:

$$(y - y_1) = m(x - x_1)$$

where the slope  $m$  is the derivative evaluated at that point:

$$m = \left[ \frac{dy}{dx} \right]_{(x_1, y_1)}$$

### Worked example — tangent to $y = e^x$ at $(1, 1)$

Find the equation of the tangent to  $y = e^x$  at the point  $(1, 1)$ .

$$(y - 1) = m(x - 1)$$

Find the slope  $m$  by evaluating the derivative at  $(1, 1)$ :

$$m = \left[ \frac{dy}{dx} \right]_{(1,1)} = \left[ \frac{d}{dx} e^x \right]_{(1,1)} = [e^x]_{(1,1)} = e^1 = e$$

Substitute  $m = e$  back into the point-slope equation and simplify:

$$(y - 1) = e(x - 1) \Rightarrow y = ex - e + 1 \Rightarrow y = ex + (1 - e)$$

**Result:** the tangent to  $y = e^x$  at  $(1, 1)$  is  $y = ex + (1 - e)$ .

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